# Yonghao Yu

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## Biography

I am a third-year Ph.D. student from Yamasaki Lab, the University of Tokyo. My recent research mainly focuses on computer vision, including **image, video and 3D generation**.

## Education

**The University of Tokyo** — Japan
*Ph.D. Student, Graduate School of Information Science and Technology, Yamasaki Laboratory* — *Oct. 2024 – Sep. 2027*

**Waseda University** — Japan
*Master of Engineering, Graduate School of Information, Production and Systems* — *Oct. 2022 – Sep. 2024*

**Sichuan University** — China
*Bachelor of Engineering in Cyber Science and Engineering* — *Sep. 2019 – Jun. 2023*

## Experience

**Research Intern**
*Alaya Lab* — *Sep. 2026 – Present*

- Conduct research on 3D generation and programmable worlds.
- First author of MeshAxiom, an agent-native 3D foundation stack; co-author of WorldSculpt, KaiNinja and EditHero.

**Research Assistant**
*The University of Tokyo* — *Oct. 2024 – Present*

- Conduct research on large-scale visual generative modeling under the Collaborative Industry Program.
- Optimize multi-node distributed training on A100/H100 clusters for visual generative modeling.

## Publications

**Video-Mirai: Autoregressive Video Diffusion Models Need Foresight**
***Yonghao Yu**, Lang Huang, Runyi Li, Zerun Wang, Toshihiko Yamasaki*

- In Advances in Neural Information Processing Systems (NeurIPS), 2026.
- Proposed Video-Mirai, a training-only foresight objective that lets future segments supervise the current causal state of autoregressive video diffusion models, improving VBench Total over Causal-Forcing and consistency beyond the training horizon with zero inference overhead.

**Mirai: Autoregressive Visual Generation Needs Foresight**
***Yonghao Yu**, Lang Huang, Zerun Wang, Runyi Li, Toshihiko Yamasaki*

- In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
- Proposed Mirai, a general foresight-based training framework for visual autoregressive models without architecture changes or inference overhead. Achieving up to 10× faster convergence and reduced ImageNet FID.

**BoostDream: Efficient Refining for High-Quality Text-to-3D Generation from Multi-View Diffusion**
***Yonghao Yu**, Shunan Zhu, Huai Qin, Haorui Li*

- In Proceedings of the 33rd International Joint Conference on Artificial Intelligence, 2024.
- Developed BoostDream, an efficient 3D refinement method integrating feed-forward generation and multi-view SDS, significantly improving quality and efficiency in text-to-3D synthesis.

**Spotlight on Video Piracy Websites: Familial Analysis Based on Multidimensional Features**
*Chenlin Wang, **Yonghao Yu**, Ao Pu, Fan Shi, Cheng Huang*

- In Proceedings of the 15th International Conference on Knowledge Science, Engineering and Management, 2022.
- Proposed a proactive detection and clustering framework for video piracy websites, achieving 96.2% detection accuracy and enabling comprehensive family-level analysis across layout, content, and infrastructure features.

## Projects

**MeshAxiom: An Agent-Native 3D Foundation Stack for Programmable Worlds**
***Yonghao Yu**, Ruihan Yu, Sho Kuno, Muyao Niu, Yu-Ju Tsai, Zheng-Hui Huang, Fengbo Lan, Guixu Lin, Runyi Li, Chuanhao Li, Ming-Hsuan Yang, Kaipeng Zhang, Zhixiang Wang*

- Open-source release with Alaya Lab, 2026: project page, code and MeshAxiom-Base model weights.
- Developed MeshAxiom-Base, which generates compact explicit meshes from one image and a requested face budget, with file-based tools that let agents inspect, edit and reuse the meshes as modules in scene programs.

**Logit Poisoning Attack in Distillation-based Federated Learning and its Countermeasures**
***Yonghao Yu**, Shunan Zhu, Jinglu Hu*

- arXiv:2401.17746, 2024.
- Proposed a two-stage logit poisoning attack targeting knowledge distillation in federated learning, and introduced an efficient defense algorithm based on benign logit distance estimation, demonstrating significant attack effectiveness and robust mitigation.

## Language and Skills

**Programming Languages**: Python, Shell
**Skills and Platforms**: Pytorch, Linux.
**Languages**: Chinese: Native, English: Fluent (TOEFL 94)
JST Support for Pioneering Research Initiated by the Next Generation (SPRING) Program, 2024-2027